of this lower-dimensional model is directly applicable to the physical three-dimensional setting, so the simpler model provides a powerful pedagogical device.

In Figure 3.3, we make use of these simplifications and draw a two-dimensional model of a spatial region of our universe. The grid-like structure indicates a convenient means of specifying positions just as a street grid gives a means of specifying locations in a city In a city, of course, one gives an address by specifying a location on the two-dimensional street grid and also giving a location in the vertical direction, such as a floor number. It is the latter information, location in the third spatial dimension, that our two-dimensional analogy suppresses for visual clarity.

In the absence of any matter or energy, Einstein envisioned that space would be flat. In our two-dimensional model, this means that the "shape" of space should be like the surface of a smooth table, as drawn in Figure 3.3. This is the image of our spatial universe commonly held for thousands of years. But what happens to space if a massive object like the sun is present? Before Einstein the answer was nothing; space (and time) were thought to provide an inert theater, merely setting the stage on which the events of the universe play themselves out. The chain of Einstein's reasoning that we have been following, however, leads to a different conclusion.

A massive body like the sun, and indeed any body, exerts a gravitational force on other objects. In the example of the terrorist bomb, we learned that gravitational forces are indistinguishable from accelerated motion. In the example of the Tornado ride, we learned that a mathematical description of accelerated motion requires the relations of curved space. These links between gravity, accelerated motion, and curved space led Einstein to the remarkable suggestion that the presence of mass, such as the sun, causes the fabric of space around it to warp, as shown in Figure 3.4. A useful, and oft-quoted, analogy is that much like a rubber membrane on which a bowling ball has been placed, the fabric of space becomes distorted due to the presence of a massive object like the sun.

According to this radical proposal, space is not merely a passive forum providing the arena for the events of the universe; rather, the shape of space responds to objects in the environment.